

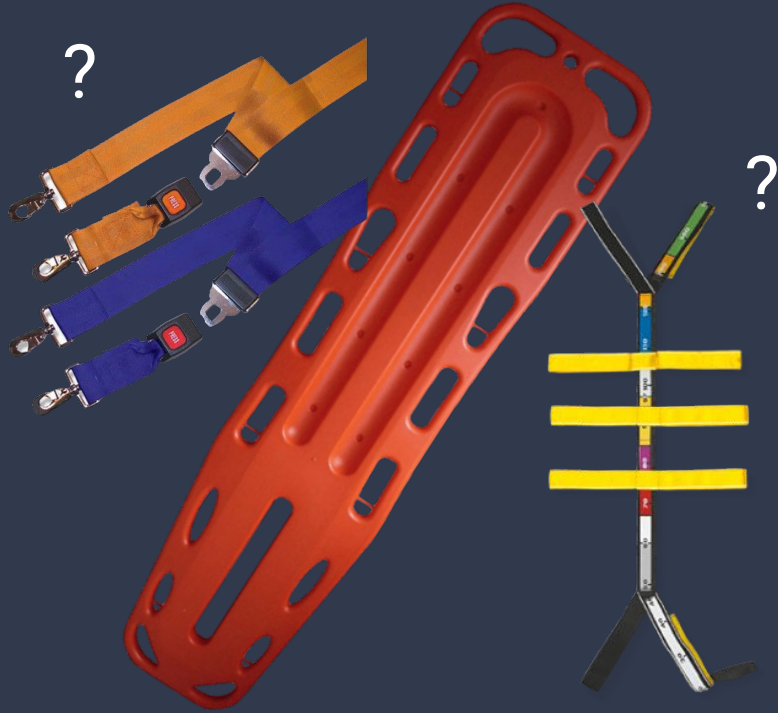
Kangaroo Belt

ENGS 21 Final Presentation

Ben Cavanagh, Sam Miller,
Oliver Shapiro, Alex Ward



Problem Statement



Rescue workers need a quick and easy way to safely secure patients to extrication spineboards.



30 Minutes

The maximum time patients can spend on a backboard before **pain** and **pressure ulcers** develop.¹

Currently, patients can spend up to **an hour** strapped in.²

1. <https://www.ncbi.nlm.nih.gov/books/NBK557714/>

2. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3691613/>



Industry Feedback

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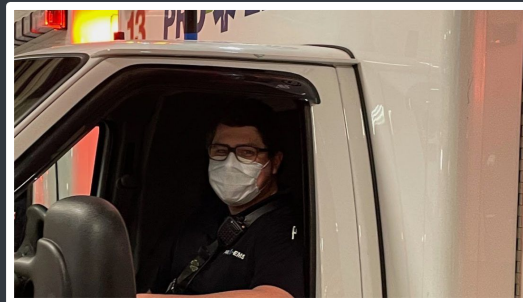
“Current spider strapping and quick clip systems are slow and inefficient. Snow already provides another barrier to efficiency when strapping patients”

Jacob Schnell, Member,
Dartmouth Ski Patrol



“There are just quite a few steps and you have to be careful to not mess up when everything is going on.”

Ayush Ajmani, Member,
Dartmouth EMS



“Sizing the straps to a patient can be difficult. We need a less time-consuming training requirement for patient securing.”

Thomas Wobby, Director of
Operations, ProEMS

Main Specifications

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Specification	Justification	Quantification	Evaluation
What requirements must ideas/solutions meet?	Why is this specification included?	What is the target value?	How can this specification be evaluated?
Time to secure patient	Time is of the essence for EMTs, and long periods of immobilization risk injury to patient.	Under 2 minutes	Timing trained EMTs on strapping patients under shoulders and across torso/hips/legs
Securing force on patients	Patients need to be immobilized during transport to prevent further injury. This will check for alignment/positioning issues.	10 Newtons minimum	Force of retraction on strap measured.
Sanitation	Straps need to be clean so as to not further sicken patients	Qualitative	Can the entire strapping system be easily sprayed and wiped down?
Legality	Should not use current technologies that are patented	Binary y/n	Continue to evaluate current market and do patent searches.
Intuitiveness: How easy is it to use?	Ensure that an EMT can use under pressure	Qualitative review by EMTs	Related to time to secure patient as well as # of steps in procedure.

Secondary Specifications

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Specification	Justification	Quantification	Evaluation
What requirements must ideas/solutions meet?	Why is this specification included?	What is the target value?	How can this specification be evaluated?
Portability	Cannot encumber application of longboard or patient movement (log roll technique)	4x6x2 inches max per strap enclosure	Volume measurement. Check it is flush with the backboard. No more than 5 separate pieces (# of quick clips) and ideally 1 piece.
(Ethical) Comfort 1: Circulation - does not constrict CSMs	Goal is to not worsen patient condition: cannot cut off blood flow.	CSMs intact	Checking tester capillary refill, sensation, and motion in extremities after 30 minutes strapped in (the maximum recommended time; patient should never be immobilized even close to this long).
(Ethical) Comfort 2: Psychological subjective feedback	Patients need to stay calm and comfortable during rescue	7-10 on a 1-10 scale of comfort. No change in breathing or heart rate.	Asking testers for 1-10 comfort scale after being strapped in for 20 minutes. Check breathing and heart rate.
Durability of strap retraction mechanism.	Straps should consistently and repeatedly be retracted into the spooling system.	Consistent force generated by rotary spring over time.	Test rotary springs over long periods of time.
Strength: weight hanging test	Patients need to be immobilized during transport to prevent spinal injury. This will check for strength and durability issues.	5500 Newtons - US seatbelt standard divided by 4	Newton meter with weights hanging from belt. Instron machine. Test both webbing, strap lock mechanism, and board connectors to determine likely failure points.

Talking to EMS professionals and playing around with seat belts and straps, we came to a general idea on how we want to solve the issue



1. Retracting Seat Belt System
2. Snug-Fit Lock



How would they differ from a car's belt system?

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Similarities

- Retracting mechanism
- A strong lock to restrain patients

Key Differences

- Locking needs to be on demand vs. force activated
- Portability and modularity - ease of connection with board



1. Retracting System
2. Clamping System
3. Locking Mechanism



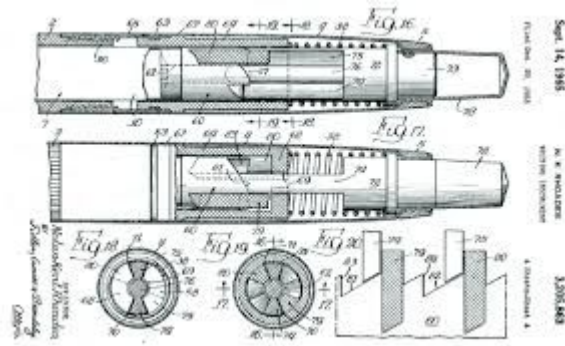
Inspiration

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Hip binder



Pen clicking system



Chip clip



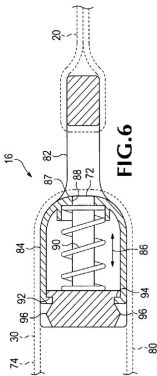
Regular strap



Cam Cleat

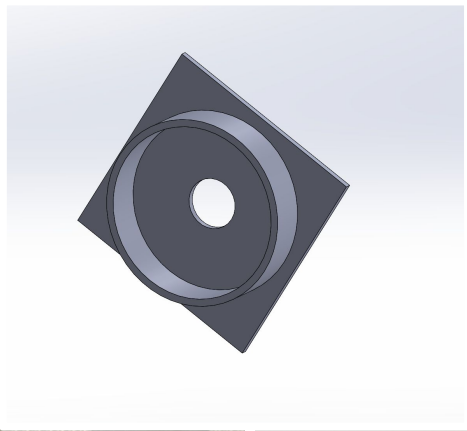
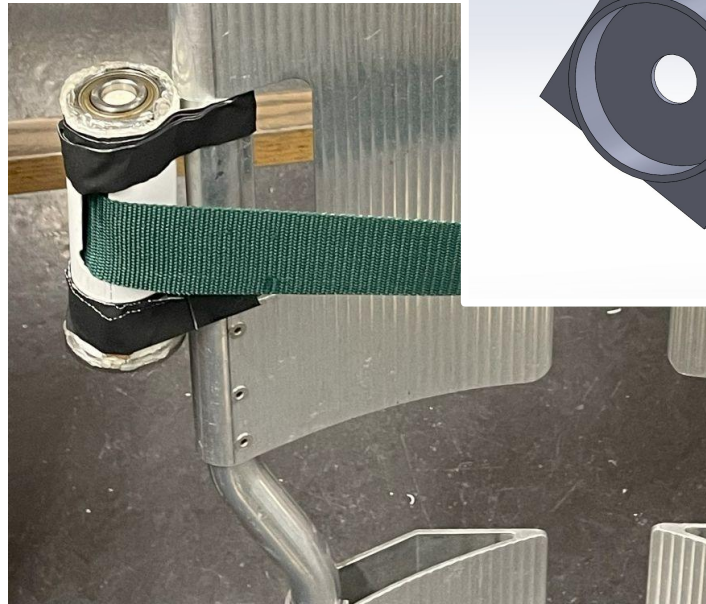


This connection



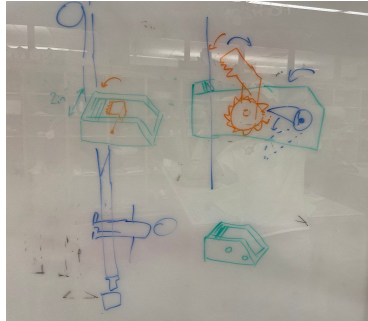
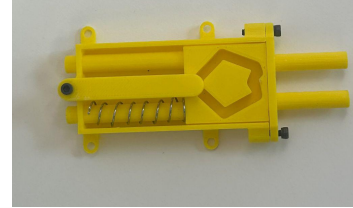
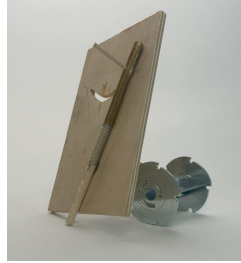
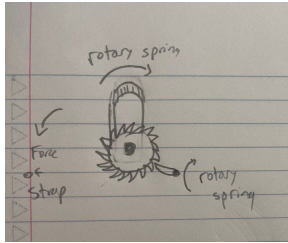
Strap Tensioning Prototype

10



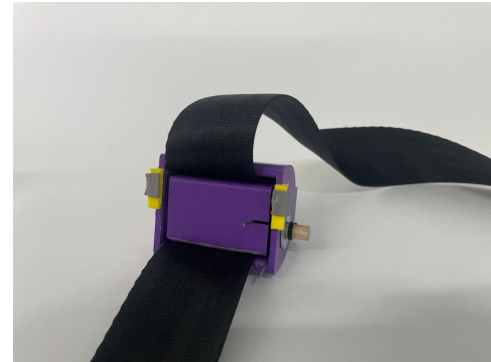
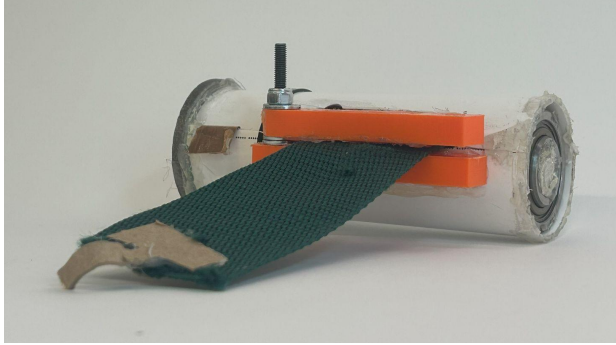
Locking Mechanism Prototypes

11



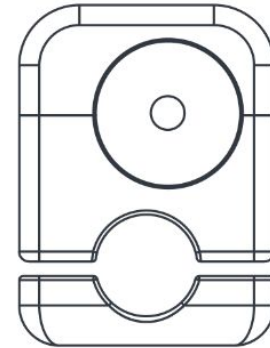
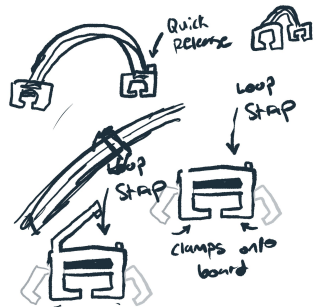
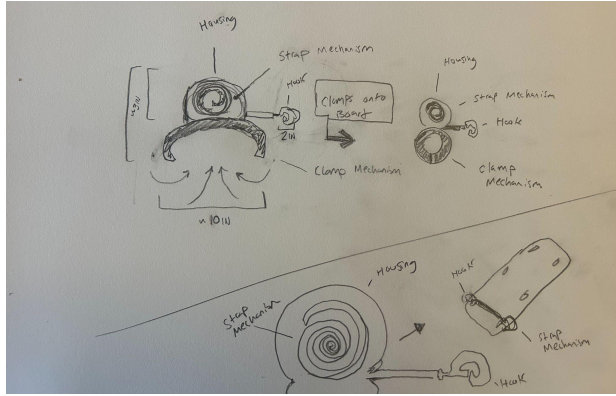
Locking Mechanism Prototypes (cont.)

12



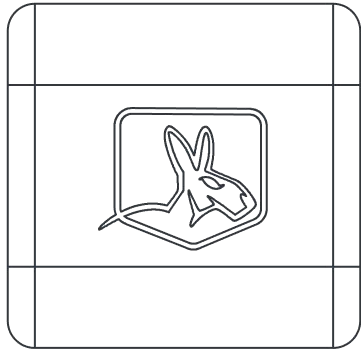
Board Clamping Prototypes

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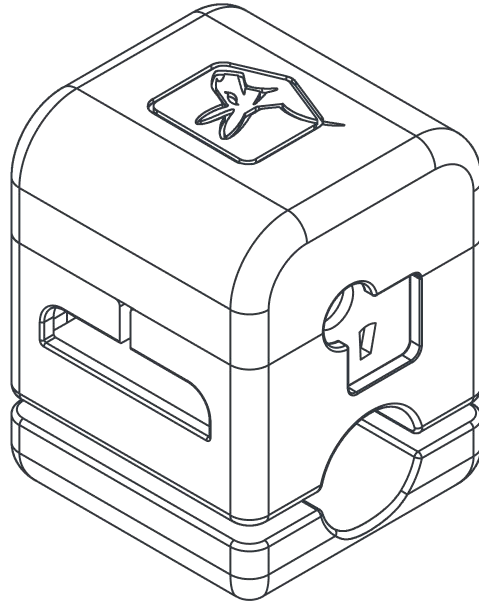


Final Prototype Renderings - Full Case

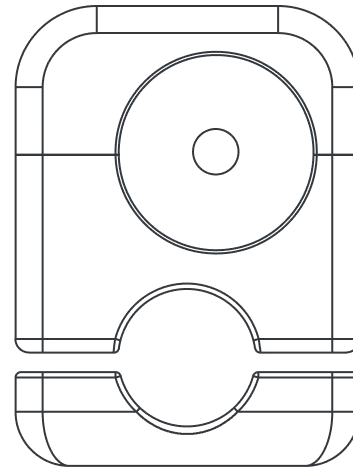
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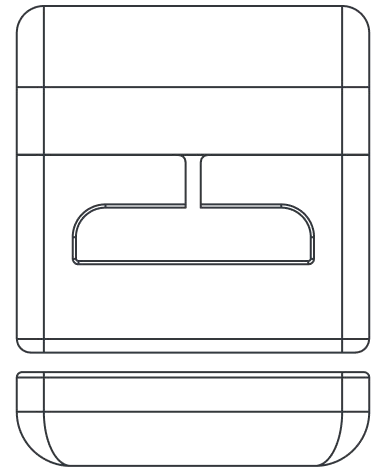
Top View



Isometric View



Side View 1/2

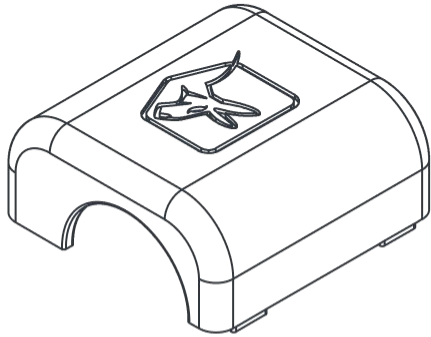


Side View 2/2

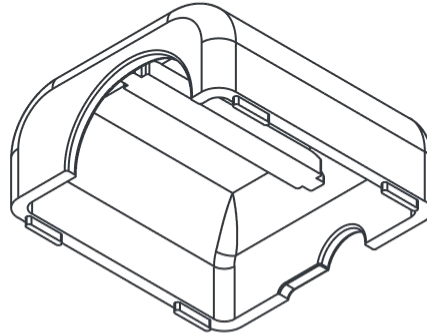


Final Prototype Renderings - Top of Case

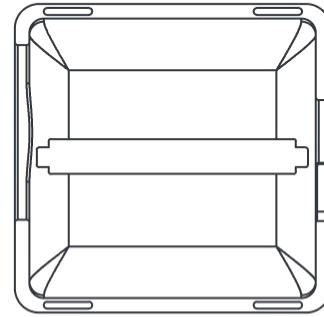
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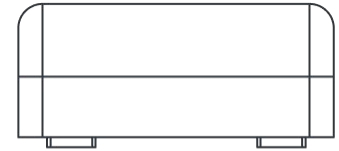
Isometric View 1/2



Isometric View 2/2



Bottom View

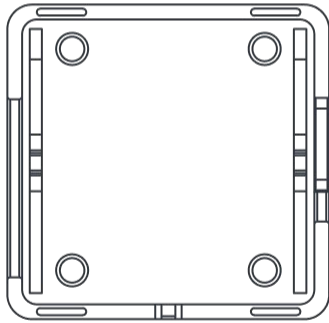


Side View

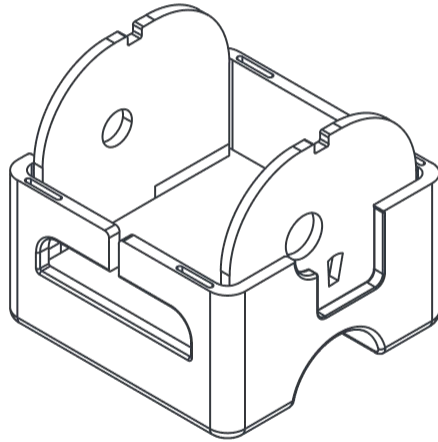


Final Prototype Renderings - Case Body

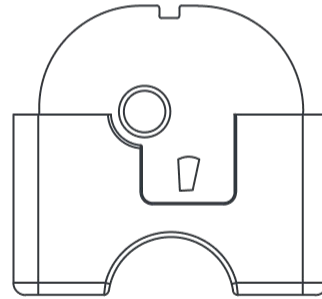
16



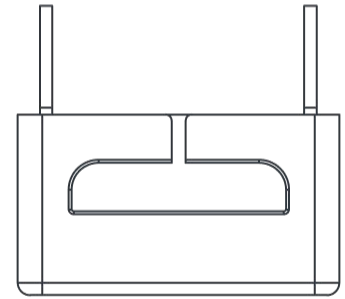
Top View



Isometric View



Side View

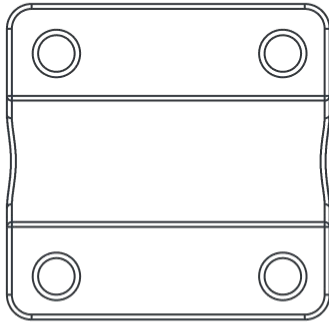


Front View

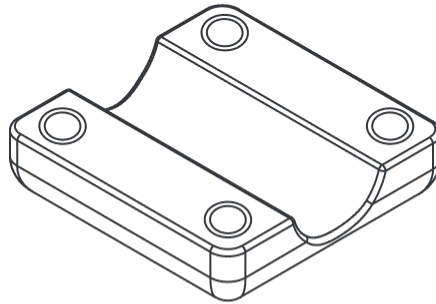


Final Prototype Renderings - Bottom of Case

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Top View



Isometric View



Side View

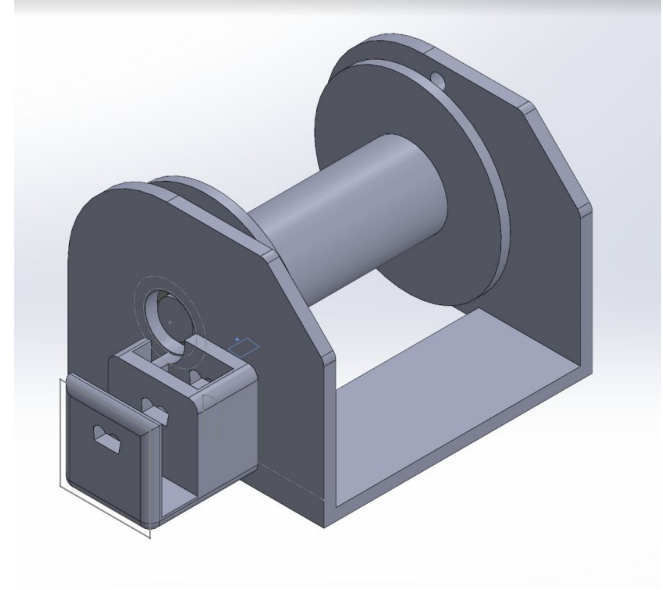
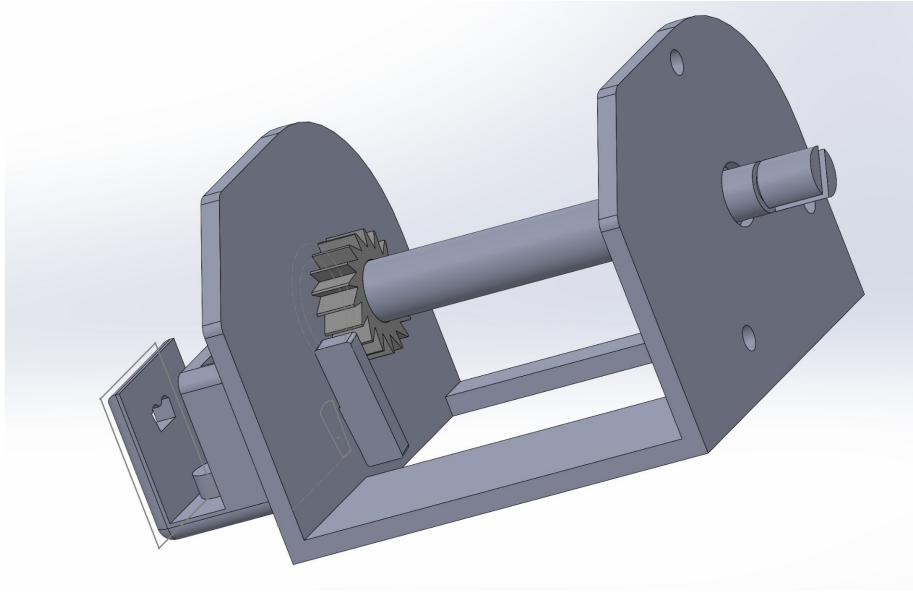


Front View



Final Prototype Renderings - Locking

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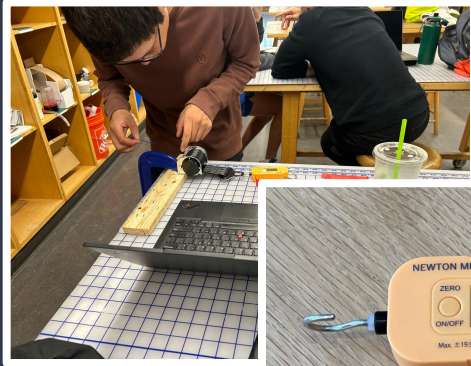


Final Prototype - Printed and Assembled

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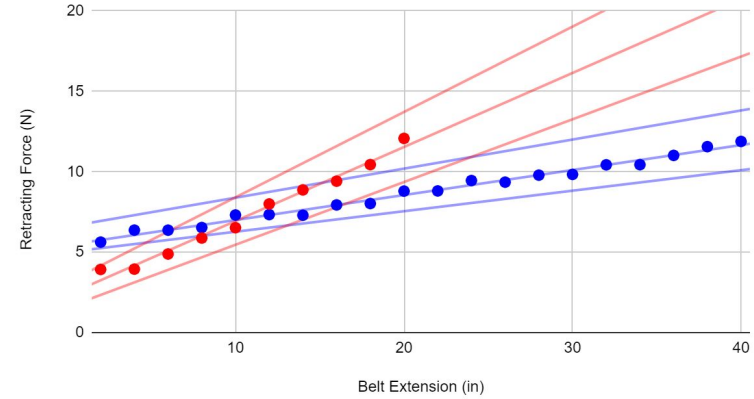


Benchmark Testing



$$\sigma = \frac{F}{A}$$

Average Retracting Force (N) vs Belt Extension (in)



	Spring Constant (N/m)	2SD
Benchmark Seatbelt	6.02	0.26
Prototype Rotary Spring	17.3	3.23

Backboard Testing: Need **10-15N of tension** for a “snug” fit.

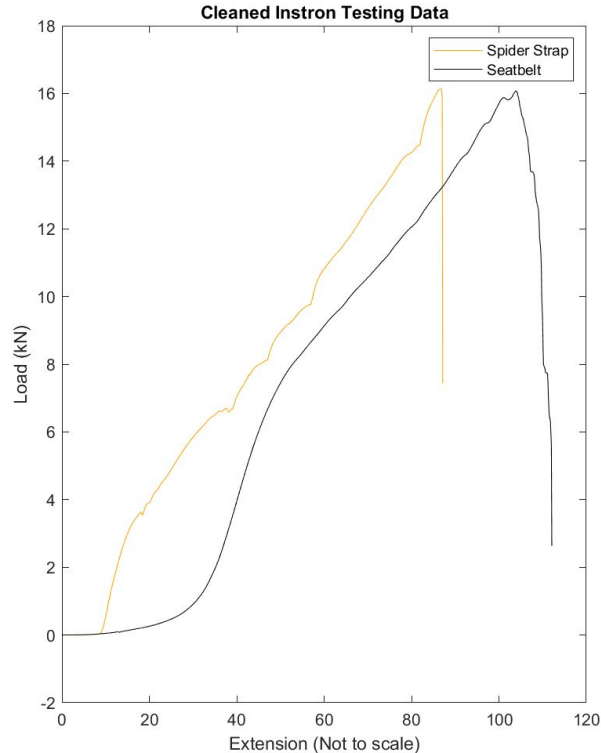
Backboard Time: Average 6 minutes.

Tensile stress: ~23 kPa



Testing: Instron Machine

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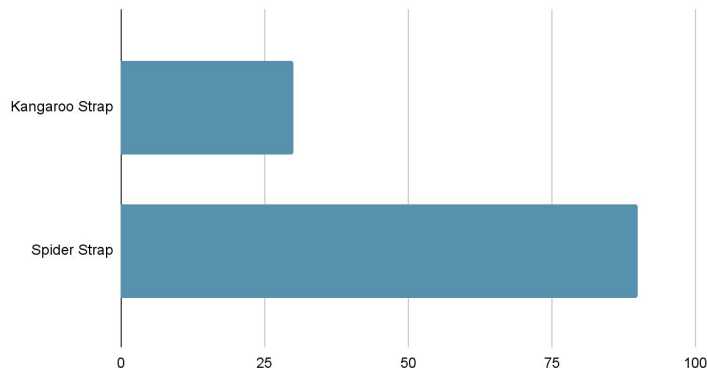
- Pulling speed of 75 mm/min based on ASTM D6775-13 Standard (Breaking Strength of Textile Webbing).
- Maximum of 16.14kN (seatbelt) and 16.07kN (spider strap)
- Caution in comparing exact test results but takeaway that stitching is the main weak point.



Total Strapping Time:

Kangaroo Belt: 30 seconds

Spider Strap: 90 seconds



Security Benchmarks:

- Lock triggers at 15N of force
- Retractor spring pulls with 10-12N of force at typical extensions.
- Benchmark: Need **10-15N of tension** for a "snug" fit.

Testing:

- Performed in EMS Practical Exam Context
- Procedure performed by 5 trained EMS technicians
- **Partial** patient immobilization: production model would have 4 straps



Testing: Dartmouth EMS Review

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What they like:

1. Quick
2. Portable
3. Simple

What they didn't like:

1. Set Placement of System

Future Goals:

1. Downsizing Clamp Connector
2. Integration of Automatic Lock
3. Modular Placement of Device



Mono-Material Parts	Manufacturing and Distribution	Lifetime
Nylon or Polyester • Sheds microplastics if machine washed Aluminum or Steel • Highly recyclable Hard Plastics • Typically high-density polyethylene (HPDE) plastic, which can be difficult to recycle.	Logistics • Simple components to minimize long-distance sourcing • No specialized manufacturing required. Sales • Sell in bulk to groups of EMS services in a given area.	Use • Reusable • Access points to spooler and lock mechanism allow for easy cleaning. • Eases patient/EMT stress during use when compared to competing products End of Life • Prioritize durability, safety, and low turnover. • Customer program to incentivize trading in old belts (in-house recycling)

Business Plan

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Key Partners



EMS Groups

EMS Training Services

Ski Patrol Groups and other
Outdoor Emergency Care
Services

Dartmouth EMS and Ski
Patrol

Customer Segments



EMS purchasers are users.
Resale/renting/public use not
intended.

Key Resources



Aluminum
Polyester Webbing
Plastics

Medical -grade factory with
additional hygiene quality
control.

Value Proposition



Lower **training cost**/time for
EMS services.

Lower-risk, quicker, and **more
consistent extraction**
process.

Lower average time for
patients to be strapped onto
a backboard due to more
efficient immobilization.

**Easier storage and
organization.**

Customer Relationships



Demonstration or training
services.

Strong customer support to
encourage replacement
purchases (possibly yearly).

Channels

Word-of-mouth or free trials.

Partnerships with medical
companies adjacent to this
niche.

Sell in bulk to EMS groups.

Key Activities



Strength and reliability testing

Adherence or positioning
relative to existing standards
or medical grades.

**Development of training
guides.**

Cost Structure



Medical premium enables some freedom in pricing structure/lower price sensitivity.

Patent: \$10,000

Material cost: \$20

Construction time: 3 hrs

Revenue Streams



Sales of RetractoBelts: **\$100 per item**

Possible recycling/repair services: \$20 + shipping

Additional cleaning/maintenance accessories: TBD

Variable Costs

- Polyester, stainless steel, HDPE plastic
- Machining, metal cutting and folding, assembly wages

Manufacturing Cost/4-Belt Pack: \$50.00

Fixed Costs

- Factory rent and upkeep
- Advertising and training sessions
- Salaried employees

Annual Fixed Costs: \$150,000

Market Size - 100,000

25,000 collegiate and public EMS agencies
Average 4 purchases per year

Break-Even Sale Volume

Price/Belt	Volume	Market Share
\$100	3000	3.0%
\$150	1500	1.5%
\$180	1154	1.2%

Price Insensitivity

EMS Agencies, especially in HCOL, high-traffic areas, typically spend large amounts of money to prioritize efficiency and patient safety .



Acknowledgements

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Thanks to the following people for wonderful help and moral support throughout the term:

Kevin Baron (**our TI**)

PJ Griffiths (**our TA**)

Andrew Wilson

Danny DeNauw

Thomas Wobby of ProEMS

Jacob Schnell + the Dartmouth Ski Patrol

(they lent us 2 backboards and straps for the entire term!)

And our user testers:

Thomas Fenaroli

Elaina Lee

Tara Karim

Ismail Habibi

Ayush Ajmani & Dartmouth EMS

+ others!



Thank you!

Appendix I - Business Plan Details

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Variable Costs		
Polyester Webbing	\$8.75	\$0.70/foot
Steel	\$2.00	\$0.65/lb
HDPE	\$3.00	\$1/lb
Injection Molding	\$6.25	0.25 hours at \$25/hr; can be done in large volume
Steel Cutting and Folding	\$10.00	0.5 hours at \$20/hr; includes machining and stamping (no welding)
Assembly	\$20.00	1 hour at \$20/hr; includes packaging and shipping
Total Variable Cost	\$50.00	For a set of four belts

Annual Fixed Costs	
Factory operation	\$18,000
Marketing/Advertising	\$10,000
Trade Shows	\$4,000
Insurance	\$2,000
Salaried Employees	\$110,000
Training Guides and Classes	\$6,000
Total Fixed Costs	\$150,000



Appendix II - Cleaning Process

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REFOLDING:

1. Lay strap out as if on patient.
2. Fold cross straps back on themselves.
3. Position yourself at the foot end of the strap and grasp the right and left sides of the foot strap.
4. Pick up cross straps in sequence, half in your right hand and half in your left hand, beginning with the bottom strap. Align the center strap folds toward you.
5. When entire strapping system has been picked up, fold in half and secure with the small hook and loop fastener located on the V-Strap.
6. Store near backboard or scoop stretcher until next use.

CARE:

1. Hand wash. For machine washing, put strap in a pillow case, use cool water and a mild detergent and the gentle cycle.

CAUTIONS:

1. Inspect the straps before each use.
2. Discard the strap if you observe: broken stitching, frayed or cut fabric or straps, and/or other signs of deterioration.
3. Do not overtighten to assure adequate circulation.
4. NOT INTENDED FOR VERTICAL LIFTING.
5. DO NOT PLACE CHEST STRAP OVER PATIENT'S DIAPHRAGM.

AIR DRY ONLY. DO NOT PUT IN DRYER. DO NOT AUTOCLAVE.

Spider-Clip Replacement Parts

EP-516	Red Strap
EP-517	Yellow Strap
EP-518	Green Strap
EP-519	Blue Strap
EP-520	Carry Bag
EP-521	Center Black Strap
EP-522	Shoulder Pads

Spider-Loop Replacement Parts

EP-511	Red Strap
EP-512	Yellow Strap
EP-513	Green Strap
EP-514	Blue Strap
EP-515	Carry Bag
EP-521	Center Black Strap
EP-522	Shoulder Pads



For best results in the field, practice using the Spider-Clip™ and the Spider-Loop™ until you become thoroughly familiar with it.



Explanation of Symbols	
Symbol	Explanation
	Manufacturer
	European Authorized Representative
	Refer to for Additional Information
	Caution, Consult Accompanying Documents

**Emergency Products & Research**
200 West Main Street
Kent, OH 44240-2218
1-800-322-5725

**EC REP**
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200 West Main Street
Kent, OH 44240-2218
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**REF** CAT #8-05
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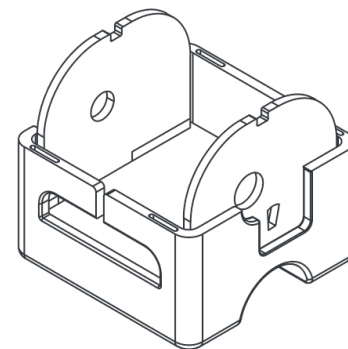
**SPIDER-CLIP™**
Immobilization Strap
With Speed Clip System
EP-500

**SPIDER-LOOP™**
Immobilization Strap
With Loop Lock System
EP-510

User Manual (EP-528)

Manufactured Exclusively By
**EMERGENCY Products + Research**
Creating Industry Standards.

- Manufacturers recommend that straps are hand washed/wiped down between uses.
- The extending belt allows for a quick wipe down.
- The internals of the system are easily accessible for deeper sanitation.



https://www.life-assist.com/Content/Docs/EPAR_Spider%20Strap%20Ops%20Manual.pdf

